ELEC3705 Assignment

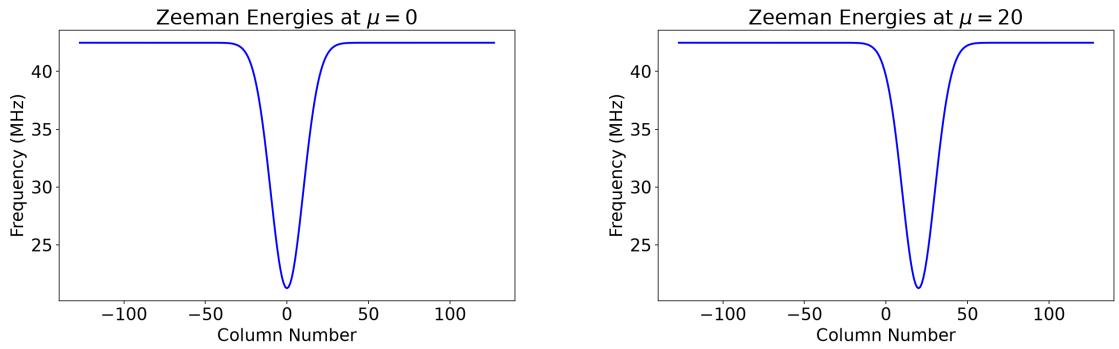
Oskar Meszaros (z5636164)

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1 Question 1.1

After loading the image, the shape of the array is 255×255.

2 Question 1.2



(a) Zeeman energy centred at column 127 ($\mu = 0$)

(b) Zeeman energy centred at column 147 ($\mu = 20$)

Figure 1: The Zeeman energies centred at column 127 and 147.

As seen above in Fig. 1, the Zeeman energies, given by

$$E_z = \gamma_n B_z,$$

are inverted Gaussians centred at the respective column number. This means that the potential energy of the hydrogen atoms are affected by the surrounding atoms as defined by the Gaussian. We can also derive this by finding the eigenvalues of the Hamiltonian, given as (for the time-independent only magnetic field),

$$\mathcal{H} = \gamma_n B_z \vec{I}_z,$$

where γ_n is the gyromagnetic ratio for the hydrogen atom, B_z is the time-independent magnetic field given by

$$B_z(c) = 1 - \frac{G(c)}{2 \cdot \max(G(c))},$$

and $\vec{I}_z$ is the z-axis nuclear spin matrix, given by $\frac{1}{2}\sigma_z$.

We can then find the eigenvalues of the Hamiltonian to be

$$\begin{aligned}\mathcal{H} &= \gamma_n B_z \frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\\ |\mathcal{H} - \lambda\mathbb{I}| &= \left|\begin{pmatrix}\frac{\gamma_n B_z}{2} - \lambda & 0\\ 0 & -\frac{\gamma_n B_z}{2} - \lambda\end{pmatrix}\right|\\ \lambda &= \pm\frac{\gamma_n B_z}{2}\end{aligned}$$

From this, we see that the splitting between the two eigenvalues is $\gamma_n B_z$, which is the Larmor frequency.

3 Question 1.3

Considering the case where the magnetic field is centred on the centre pixel, we have the Hamiltonian to be

$$\mathcal{H}(c) = \gamma_n\left(1 - \frac{1}{2}e^{-c^2/200}\right)\vec{I}_z$$

This means that for the centre pixel, we have the Hamiltonian to be

$$\mathcal{H}(0) = \gamma_n\left(1 - \frac{1}{2}e^{-0}\right)\vec{I}_z = \frac{1}{2}\gamma_n\vec{I}_z,$$

and for the last pixel it is

$$\mathcal{H}(127) = \gamma_n\left(1 - \frac{1}{2}e^{-127^2/200}\right)\vec{I}_z \approx \gamma_n\vec{I}_z.$$

4 Question 1.4

We now have an additional magnetic field given by

$$B_x = B_{amp}\cos 2\pi f t,$$

which is a simple linearly oscillating field, so we are able to represent it using two counter-rotating magnetic fields. Using this representation, we get the field to be

$$B_x = B_{amp}\left[\frac{e^{i2\pi f t} + e^{-i2\pi f t}}{2}\right].$$

Here, $e^{i2\pi f t}$ is the on-resonance component driving the spins, and the $e^{-i2\pi f t}$ is the off-resonance component which is detuned by -2ω, and has no effect on the spin. This gives the field to be

$$B_x = \frac{1}{2}B_{amp}e^{i2\pi f t}.$$

At resonance, the Rabi oscillation frequency is then $\frac{\omega_1}{2}$, where $\omega_1 = \gamma_n B_{amp}$ in field, and hence the Rabi frequency is

$$\omega_R = \frac{\gamma_n B_{amp}}{2}.$$